



Zero Trust WARP: tunneling with a MASQUE

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5 min read



Slipping on the MASQUE[🔗]

In June 2023, we [told you](#) that we were building a new protocol, [MASQUE](#), into WARP. MASQUE is a fascinating protocol that extends the capabilities of

[HTTP/3](#) and leverages the unique properties of the QUIC transport protocol to efficiently proxy IP and UDP traffic without sacrificing performance or privacy

At the same time, we've seen a rising demand from [Zero Trust](#) customers for features and solutions that only MASQUE can deliver. All customers want WARP traffic to look like HTTPS to avoid detection and blocking by firewalls, while a significant number of customers also require FIPS-compliant encryption. We have something good here, and it's been proven elsewhere (more on that below), so we are building MASQUE into Zero Trust WARP and will be making it available to all of our Zero Trust customers — at WARP speed!

This blog post highlights some of the key benefits our Cloudflare One customers will realize with MASQUE.

Before the MASQUE[🔗]

Cloudflare is on a mission to help build a better Internet. And it is a journey we've been on with our device client and WARP for almost five years. The precursor to WARP was the 2018 launch of [1.1.1.1](#), the Internet's fastest, privacy-first consumer DNS service. WARP was introduced in 2019 with the [announcement](#) of the 1.1.1.1 service with WARP, a high performance and secure consumer DNS and VPN solution. Then in 2020, we [introduced](#) Cloudflare's Zero Trust platform and the Zero Trust version of WARP to help any IT organization secure their environment, featuring a suite of tools we first built to protect our own IT systems. Zero Trust WARP with MASQUE is the next step in our journey.



The current state of WireGuard[↗]

[WireGuard](#) was the perfect choice for the 1.1.1.1 with WARP service in 2019. WireGuard is fast, simple, and secure. It was exactly what we needed at the time to guarantee our users' privacy, and it has met all of our expectations. If we went back in time to do it all over again, we would make the same choice.

But the other side of the simplicity coin is a certain rigidity. We find ourselves wanting to extend WireGuard to deliver more capabilities to our Zero Trust customers, but WireGuard is not easily extended. Capabilities such as better session management, advanced congestion control, or simply the ability to use FIPS-compliant cipher suites are not options within WireGuard; these capabilities would have to be added on as proprietary extensions, if it was even possible to do so.

Plus, while WireGuard is popular in VPN solutions, it is not standards-based, and therefore not treated like a first class citizen in the world of the Internet, where non-standard traffic can be blocked, sometimes intentionally, sometimes not. WireGuard uses a non-standard port, port 51820, by default. Zero Trust WARP changes this to use port 2408 for the WireGuard tunnel, but it's still a non-standard port. For our customers who control their own firewalls, this is not an issue; they simply allow that traffic. But many of the large number of public Wi-Fi locations, or the approximately 7,000 ISPs in the world, don't know anything about WireGuard and block these ports. We've also faced situations where the ISP does know what WireGuard is and blocks

it intentionally.

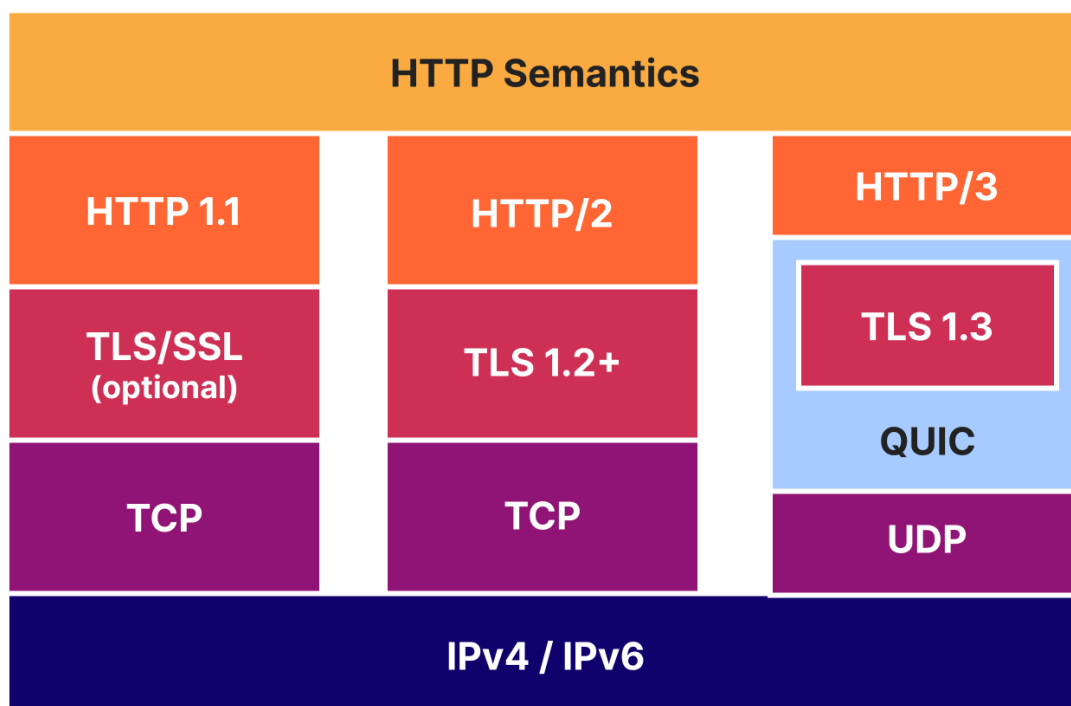
This can play havoc for roaming Zero Trust WARP users at their local coffee shop, in hotels, on planes, or other places where there are captive portals or public Wi-Fi access, and even sometimes with their local ISP. The user is expecting reliable access with Zero Trust WARP, and is frustrated when their device is blocked from connecting to Cloudflare's global network.

Now we have another proven technology — MASQUE — which uses and extends HTTP/3 and QUIC. Let's do a quick review of these to better understand why Cloudflare believes MASQUE is the future.

Unpacking the acronyms^{[↗](#)}

HTTP/3 and QUIC are among the most recent advancements in the evolution of the Internet, enabling faster, more reliable, and more secure connections to endpoints like websites and APIs. Cloudflare worked closely with industry peers through the [Internet Engineering Task Force](#) on the development of [RFC 9000](#) for QUIC and [RFC 9114](#) for HTTP/3. The technical background on the basic benefits of HTTP/3 and QUIC are reviewed in our 2019 blog post where we announced [QUIC and HTTP/3 availability](#) on Cloudflare's global network.

Most relevant for Zero Trust WARP, QUIC delivers better performance on low-latency or high packet loss networks thanks to packet coalescing and multiplexing. QUIC packets in separate contexts during the handshake can be coalesced into the same UDP datagram, thus reducing the number of receive and system interrupts. With multiplexing, QUIC can carry multiple HTTP sessions within the same UDP connection. Zero Trust WARP also benefits from QUIC's high level of privacy, with TLS 1.3 designed into the protocol.



MASQUE unlocks QUIC's potential for proxying by providing the application layer building blocks to support efficient tunneling of TCP and UDP traffic. In Zero Trust WARP, MASQUE will be used to establish a tunnel over HTTP/3, delivering the same capability as WireGuard tunneling does today. In the future, we'll be in position to add more value using MASQUE, leveraging Cloudflare's ongoing participation in the [MASQUE Working Group](#). This blog post is a good read for those interested in [digging deeper into MASQUE](#).

OK, so Cloudflare is going to use MASQUE for WARP. What does that mean to you, the Zero Trust customer?

Proven reliability at scale[🔗]

Cloudflare's network today spans more than 310 cities in over 120 countries, and interconnects with over 13,000 networks globally. HTTP/3 and QUIC were

introduced to the Cloudflare network in 2019, the HTTP/3 standard was [finalized in 2022](#), and represented about [30% of all HTTP traffic on our network in 2023](#).

We are also using MASQUE for [iCloud Private Relay](#) and other Privacy Proxy partners. The services that power these partnerships, from our Rust-based [proxy framework](#) to our open source [QUIC implementation](#), are already deployed globally in our network and have proven to be fast, resilient, and reliable.

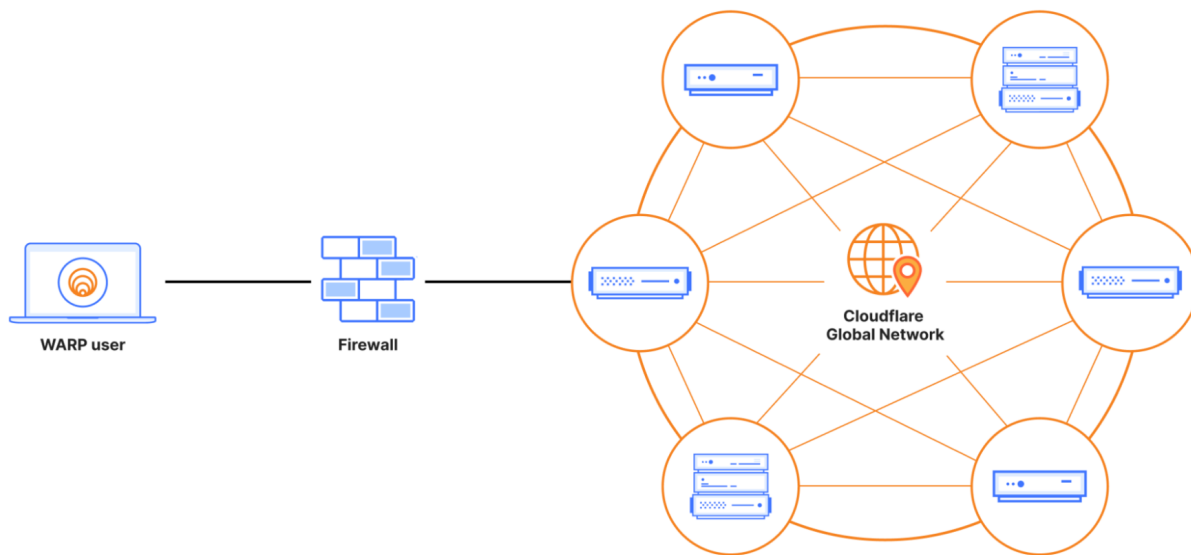
Cloudflare is already operating MASQUE, HTTP/3, and QUIC reliably at scale. So we want you, our Zero Trust WARP users and Cloudflare One customers, to benefit from that same reliability and scale.

Connect from anywhere [↗](#)

Employees need to be able to connect from anywhere that has an Internet connection. But that can be a challenge as many security engineers will configure firewalls and other networking devices to block all ports by default, and only open the most well-known and common ports. As we pointed out earlier, this can be frustrating for the roaming Zero Trust WARP user.

We want to fix that for our users, and remove that frustration. HTTP/3 and QUIC deliver the perfect solution. QUIC is carried on top of UDP ([protocol number 17](#)), while HTTP/3 uses [port 443](#) for encrypted traffic. Both of these are well known, widely used, and are very unlikely to be blocked.

We want our Zero Trust WARP users to reliably connect wherever they might be.



Compliant cipher suites^{[↗](#)}

MASQUE leverages [TLS 1.3](#) with QUIC, which provides a number of cipher suite choices. WireGuard also uses standard cipher suites. But some standards are more, let's say, standard than others.

NIST, the [National Institute of Standards and Technology](#) and part of the US Department of Commerce, does a tremendous amount of work across the technology landscape. Of interest to us is the NIST research into network security that results in [FIPS 140-2](#) and similar publications. NIST studies individual cipher suites and publishes lists of those they recommend for use, recommendations that become requirements for US Government entities. Many other customers, both government and commercial, use these same recommendations as requirements.

Our first MASQUE implementation for Zero Trust WARP will use [TLS 1.3](#) and FIPS compliant cipher suites.



How can I get Zero Trust WARP with MASQUE?



Cloudflare engineers are hard at work implementing MASQUE for the mobile apps, the desktop clients, and the Cloudflare network. Progress has been good, and we will open this up for beta testing early in the second quarter of 2024 for Cloudflare One customers. Your account team will be reaching out with participation details.

Continuing the journey with Zero Trust WARP



Cloudflare launched WARP five years ago, and we've come a long way since. This introduction of MASQUE to Zero Trust WARP is a big step, one that will immediately deliver the benefits noted above. But there will be more — we believe MASQUE opens up new opportunities to leverage the capabilities of QUIC and HTTP/3 to build innovative [Zero Trust solutions](#). And we're also continuing to work on other new capabilities for our Zero Trust customers. Cloudflare is committed to continuing our mission to help build a better Internet, one that is more private and secure, scalable, reliable, and fast. And if you would like to join us in this exciting journey, check out our [open positions](#).

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